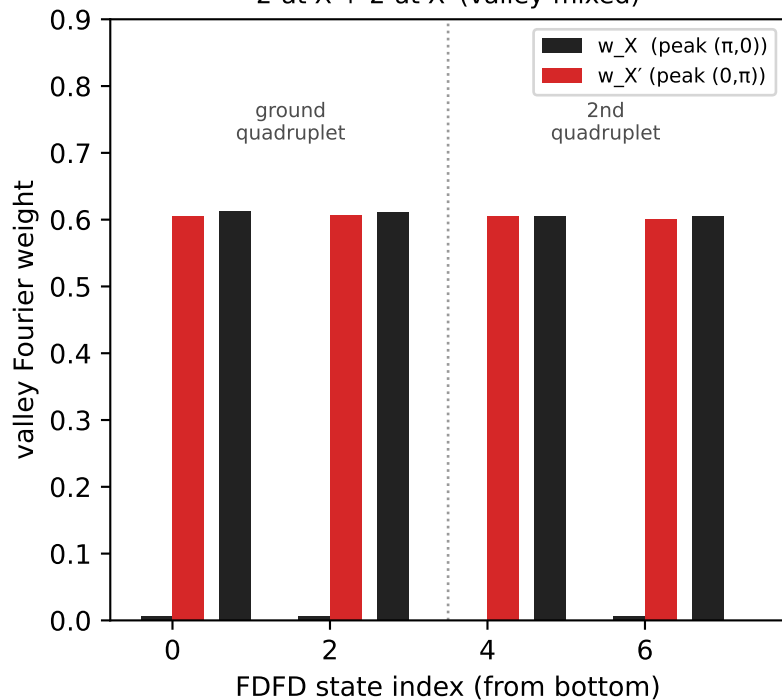
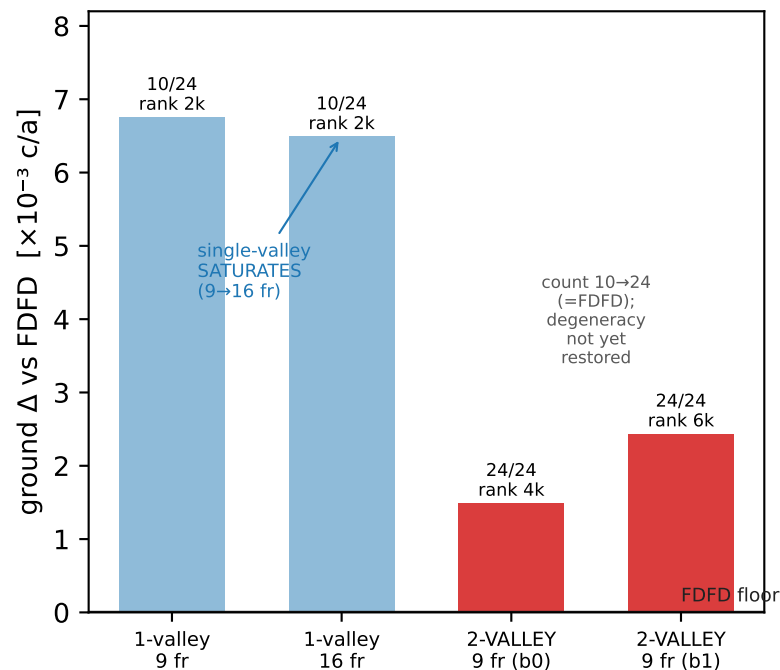


-valley ($X \oplus X'$) completion corrects §9: the missing ingredient is a 2nd momentum carrier (X'), not real-space adaptation — it lifts the plateau 2.8–4.5 $\times$ and recovers the count; full exactness needs $X \leftrightarrow X'$ coupling

A the premise: ground quadruplet is 2-at- X + 2-at- X' (valley-mixed)



B X' lifts the §9 plateau 2.8–4.5 $\times$ — it's the VALLEY, not basis size



C mechanism (fast (7,1) cell): X' -shell converges the split toward FDFD

